

PROJECT: RETAIL BUSINESS ANALYSIS

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A. PROJECT OVERVIEW

Business Context: Retail businesses generate large volumes of transactional data, but raw sales records alone do not directly answer business questions such as:

- Which products actually contribute the most profit?
- Which customers generate the highest business value?
- Which product categories combine strong demand with attractive margins?
- Which stores use their retail space efficiently?
- Are new stores creating additional demand or merely shifting sales from existing locations?

This project uses SQL to transform transactional retail data into business-oriented insights across sales, customers, products, and store operations.

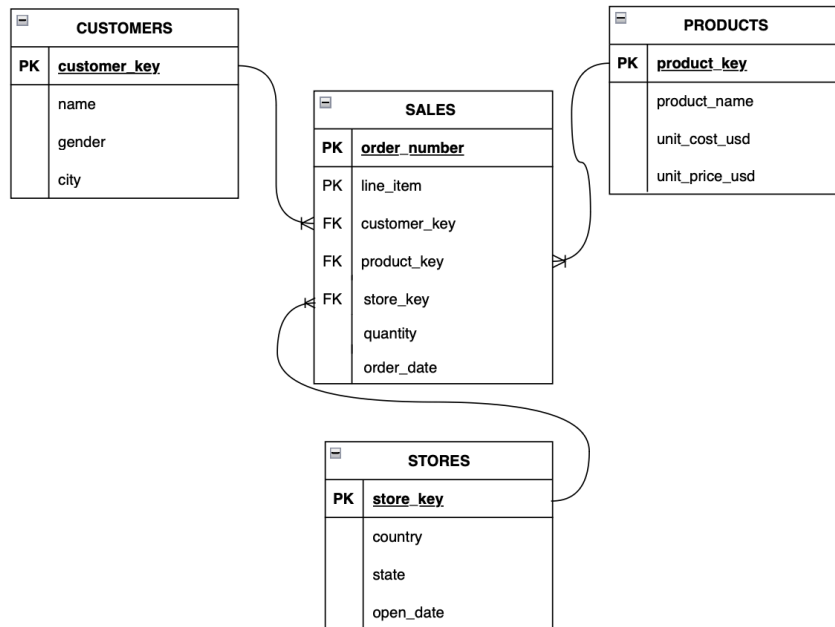
Objective: The objective is to demonstrate how SQL can be used beyond basic data retrieval to translate business questions into data-driven insights and decision support.

B. DATASET

The analysis uses a retail relational database consisting of four core tables:

Table	Role
sales	Order and transaction records
customers	Customer demographic and geographic information
products	Product, category, price, and cost information
stores	Store location, size, and opening information

Data Relationship



Because price and cost information are stored separately from sales transactions, multiple tables must be joined to calculate business KPIs such as revenue, gross profit, and customer value.

C. ANALYTICAL FRAMEWORK

The project is structured around five business areas, moving from descriptive business performance to more strategic analytical questions.

1. Sales Performance (Evaluate overall business activity and establish key sales benchmarks through order volume and revenue analysis)
2. Customer and Market Analysis (Understand geographic customer concentration and identify high-value customers across different markets)
3. Product and Profitability Analysis (Evaluate product assortment, sales volume, margin structure)
4. Store & Operations Analysis (Assess retail network coverage, delivery performance, and store productivity using normalized operational KPIs)
5. Advanced Business Analysis (Investigate potential store cannibalization to support expansion decisions)

D. DETAILED ANALYSIS

1. Sales Performance

1.1. Business Activity in 2020

- **Business Question:** How much purchasing activity did the retail business generate during 2020?
- **Result:** 4,650 distinct orders were recorded in 2020.
- **Business Interpretation:** The result establishes a baseline for transaction activity during the year.

- SQL Concepts Applied: COUNT(DISTINCT), Date Filtering, Transaction-Level Aggregation

1.2. Monthly Revenue Benchmark

- Business Question: How much revenue was generated across the business in December 2020?
- Result: December 2020 Revenue: \$651,526.44
- Business Interpretation: December generated approximately \$651.5K in sales revenue, providing a useful monthly benchmark for commercial activity. Because selling price is stored in the products table rather than the transaction table, the calculation required joining sales and products through product_key.
- SQL Concepts Applied: INNER JOIN, SUM(), Calculated Metrics, Date Filtering

2. Customer and Market Analysis

2.1. Geographic Customer Concentration

- Business Question: Which cities contain the largest customer populations?

Rank	City	State / Province	Country	Customers
1	Toronto	Ontario	Canada	203
2	New York	New York	United States	130
3	Los Angeles	California	United States	119
4	Montreal	Quebec	Canada	97
5	Chicago	Illinois	United States	90
6	Houston	Texas	United States	84
7	Calgary	Alberta	Canada	70
8	Dallas	Texas	United States	67
9	Vancouver	British Columbia	Canada	59
10	Philadelphia	Pennsylvania	United States	53

- Key Findings:
- + Customer concentration is particularly strong in major North American metropolitan areas.
- + Toronto represents the largest customer market with 203 customers, approximately 56% more customers than New York, the second-ranked city.
- + The top ten consists exclusively of U.S. and Canadian cities: 6 U.S. cities, 4 Canadian cities
- Business Implication: This suggests substantial customer concentration in North America within the dataset. Markets with large customer populations may represent important areas for localized marketing, customer engagement, distribution planning, and market-level performance monitoring. However, customer count alone does not indicate market value; cities with fewer customers may still generate higher revenue per customer.

- SQL Concepts Applied: TOP, GROUP BY, COUNT(), Multi-column Geographic Grouping, Ranking.

2.2. High-Value Customers by Country

- Business Question: Who was the highest-spending customer in each country during 2020?

Country	Highest-Value Customer	2020 Spending
United States	Mie Huus	\$33,275.47
Canada	Zane Belgrave	\$20,644.89
United Kingdom	Dominic Banks	\$19,371.06
France	Alice Lafond	\$18,659.88
Germany	Daniel Kaestner	\$17,931.00
Australia	Makayla Hassall	\$16,778.46
Italy	Gaspare Trevisan	\$13,831.89
Netherlands	Polat Klarenbeek	\$13,369.52

- Key Findings: The highest-spending U.S. customer generated approximately \$33.3K in 2020. This is around 61% higher than the highest-spending customer in Canada, which ranked second among the country-level VIP customers. The result demonstrates considerable variation in top-customer value across markets.
- Business Implication: Country-level VIP identification can support premium customer programs, targeted retention campaigns, personalized offers, high-value customer engagement, and CRM prioritization.
- SQL Concepts Applied: Multiple JOINS, CTE, Aggregation, ROW_NUMBER() · PARTITION BY

3. Product and Profitability Analysis

3.1. Product Assortment Structure

- Business Question: How is the product catalog distributed across categories?

Category	Number of SKUs
Home Appliances	661
Computers	606
Cameras and camcorders	372
Cell phones	285
TV and Video	222

Games and Toys	166
Audio	115
Music, Movies and Audio Books	90

- Key Finding: The product catalog contains 2,517 SKUs across eight categories. Home Appliances and Computers alone contain approximately 50.3% of the entire assortment.
- Business Interpretation: The catalog is therefore heavily concentrated in these two product categories. However, a broad assortment does not necessarily imply strong commercial performance, so category breadth should be interpreted together with demand and margin.
- SQL Concepts Applied: GROUP BY, COUNT(), Category-Level Aggregation

3.2. Best-Selling Products Within Each Category

- Business Question: Which products generate the highest unit sales within each category?
- Solution: Products were first aggregated by total quantity sold and then ranked within their respective categories. This avoids comparing fundamentally different categories solely on overall volume.
- Example: Audio category

Product	Units Sold
WWI 1GB Digital Voice Recorder Pen E100 Black	431
WWI 1GB Digital Voice Recorder Pen E100 Red	388
WWI 4GB Video Recording Pen X200 Yellow	370
WWI 4GB Video Recording Pen X200 Black	366
WWI Wireless Bluetooth Stereo Headphones M270 Black	359

- Key Finding: Within Audio, sales are distributed across several leading SKUs rather than being dominated by a single product. The top-ranked item sold 431 units, while the fifth-ranked item still sold 359 units. This relatively narrow gap suggests that demand among the leading Audio products is reasonably balanced.
- Business Implication: Category-level rankings can support replenishment prioritization, assortment planning, safety-stock decisions, and merchandising.
- SQL Concepts Applied: CTE, SUM(), DENSE_RANK(), PARTITION BY, Category-Level Ranking

3.3. Product Margin Structure

- Business Question: Which product subcategories offer the strongest average gross margins? Only subcategories containing at least ten products were included in order to avoid drawing conclusions from very small assortment groups.
- Results: Leading Subcategories

Subcategory	Approx. Average Margin
Digital SLR Cameras	60.44%

Digital Cameras	57.39%
Projectors & Screens	57.33%
Movie DVD	57.24%
Monitors	56.71%
Printers, Scanners & Fax	56.38%
Camcorders	56.26%

- **Key Finding:** Digital SLR Cameras show the highest average product-level gross margin at approximately 60.4%. Several other electronics-related subcategories cluster around the 55–57% range.
- **Business Interpretation:** This metric represents the average product-level margin percentage and is not weighted by actual units sold. Therefore, it reflects margin structure rather than realized profitability. In addition, the result also illustrates why revenue and sales volume should not be treated as equivalent to profitability. A category can sell heavily but produce weak margins, sell moderately but generate strong margins, or combine both strong demand and attractive unit economics. This information can therefore support pricing, assortment, and promotion decisions.
- **SQL Concepts Applied:** AVG(), HAVING, NULLIF(), Calculated Margin, Subcategory Aggregation

4. Store and Operation Analysis

4.1. Retail Network Footprint

- **Business Question:** How is the store network distributed across countries?

Market	Store Records
United States	24
Germany	9
France	7
United Kingdom	7
Australia	6
Canada	5
Netherlands	5
Italy	3
Online	1

- Key Finding: The dataset contains 67 store records, including one record representing the online channel. The United States has 24 store records, representing the largest physical retail presence in the dataset. This accounts for roughly 36% of all store records.
- Business Interpretation: The distribution shows that the company's physical footprint is not equally spread across markets.
- SQL Concepts Applied: COUNT(DISTINCT), GROUP BY, Market-Level Aggregation

4.2. International Delivery Performance

- Business Question: How long does delivery take on average across countries?

Country	Delivered Orders	Avg. Delivery Days
Italy	211	4.68
Canada	512	4.67
United Kingdom	637	4.64
France	133	4.53
United States	3,068	4.52
Australia	281	4.44
Netherlands	200	4.35
Germany	538	4.35

- Key Finding: Average international delivery performance is relatively consistent. The observed range is approximately: 4.35–4.68 days. The difference between the fastest and slowest country averages is only: 0.33 day, or roughly 8 hours.
- Business Interpretation: Although Italy and Canada show slightly longer average delivery times, the differences are relatively small.
- SQL Concepts Applied: CTE, DATEDIFF(), AVG(), Distinct Order Aggregation, Geographic Comparison

4.3. Store Productivity

- Business Question: Are larger stores necessarily more productive?
- Example: Australia

Store	Revenue per m ²	Country Quartile
Store 1	100.08	1
Store 5	57.77	2
Store 6	55.57	3
Store 4	17.47	4

- **Key findings:** Store productivity can differ substantially even within the same market. In detail, in Australia, Store 1 generates approximately \$100.08 per m². Store 4 generates approximately \$17.47 per m². It means that the highest-performing store generates approximately 5.7 times more revenue per square meter than the lowest-performing store. Canada also shows meaningful variation, with Stores 10, 9, and 8 generating approximately \$162.17, 129.76, and 76.38 in revenue per m², respectively.
- **Business Implication:** Revenue per square meter can support store benchmarking, space optimization, expansion planning, and identification of underperforming locations.
- **SQL Concepts Applied:** Multiple CTEs, Revenue Calculation, Normalized KPI, NTILE(), Country-Level Benchmarking

5. Advanced Business Analysis

5.1. Potential Store Cannibalization

- **Business Question:** When a new store opens in an existing market, do incumbent stores experience lower revenue afterward?
- **Solution:** For each incumbent/new-store pair within the same country, revenue at the existing store was compared across the six months before the new store opened, and the six months after opening. Cases where post-opening revenue declined by more than 15% were flagged for further investigation.
- **Selected Results:**

Existing Store	New Store	Country	Revenue Before	Revenue After	Change
19	26	Germany	\$96,124	\$33,267	-65.39%
21	26	Germany	\$139,315	\$70,006	-49.75%
56	49	United States	\$186,585	\$125,598	-32.69%
20	21	Germany	\$81,828	\$55,457	-32.23%
23	26	Germany	\$132,438	\$89,891	-32.13%

- **Key Finding:** Several incumbent stores experienced substantial revenue declines after another store opened in the same country. The most pronounced example in the results is Germany Store 19 => Store 26 opening: -65.39%. This makes these locations reasonable candidates for further cannibalization analysis. Moreover, these results indicate potential cannibalization signals rather than causal proof. Because the dataset does not include precise geographic distance or customer catchment information, same-country store pairs are used only as a screening mechanism.
- **Business Implication:** The flagged stores provide a prioritized investigation list for expansion planning.
- **SQL Concepts Applied:** Multiple CTEs, Self-Comparison, Date Windows, Conditional Aggregation, Before/After Analysis

E. KEY TAKEAWAYS

1. Key Findings

- **Sales**
 - The company recorded 4,650 orders in 2020, while December alone generated approximately \$651.5K in revenue.
 - These values provide useful activity benchmarks, although trend comparisons are required before judging performance direction.

- *Customers*
 - Customer concentration is strongest in major North American cities, with Toronto representing the largest city-level customer base.
 - High-value customer spending also differs substantially across markets, demonstrating that customer volume and customer value should be evaluated separately.
- *Products*
 - The catalog contains 2,517 SKUs, with approximately half concentrated in Home Appliances and Computers.
 - Product margins vary meaningfully by subcategory, highlighting the need to evaluate assortment decisions using both sales volume and profitability.
- *Operations*
 - International average delivery times are relatively stable, with only around 0.33 day separating the highest and lowest country averages.
 - This suggests limited evidence of major country-level delivery inefficiency based on average lead time alone.
- *Stores*
 - Store productivity varies much more strongly than delivery performance.
 - In some markets, revenue per square meter differs several-fold between locations, suggesting meaningful opportunities for store-level performance investigation.
- *Strategic Analysis*
 - Several stores show substantial post-opening revenue declines when additional locations enter the same country, creating potential cannibalization signals. However, these patterns require geographic and causal validation.
- 2. Business Recommendations**
 - *Evaluate product performance using both demand and profitability:* High unit sales should be combined with realized gross margin before inventory or promotion decisions are made.
 - *Investigate low-productivity stores:* Revenue per square meter provides a useful screening KPI for identifying locations requiring deeper operational review.
 - *Validate potential cannibalization before further expansion:* Store-opening decisions should incorporate proximity, customer overlap, and incumbent-store performance rather than evaluating new locations independently.
- 3. Limitations and Next Steps**

The current analysis has several limitations:

- Store cannibalization is screened using market and opening-date relationships but lacks geographic distance and customer-overlap data.
- Monthly and yearly revenue trends should be incorporated for stronger temporal performance analysis.

Future enhancements may include sales-weighted profitability, validated customer retention, geographic cannibalization analysis, and interactive BI visualization.